

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belgaum-590014



An Internship Report

On

PYTHON AND DATA ANALYTICS

Submitted in Partial fulfillment of the Requirements for the VIII Semester of the Degree of

Bachelor of Engineering

In

Computer Science & Engineering

By

V Sudeep

1CE22CS170



CITY ENGINEERING COLLEGE

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



CERTIFICATE

Certified that the Internship work entitled “**PYTHON AND DATA ANALYTICS**” has been carried out by V SUDEEP (1CE22CS170) a bonafide student of City Engineering College in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visveswaraya Technological University, Belgaum during the year **2025-2026**. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the Report deposited in the departmental library. The Internship has been completed and it satisfies the academic requirements as per VTU norms with respect of Internship Work prescribed for the said Degree.

**Mr. Mahadeva
Prasad H M**
Internship Guide

Mr. Satheesh Kumar G
Internship Coordinator

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Dr. Thippeswamy H N
Principal, CEC

Name of the Examiners

External Viva

Signature with date

1.

2.

Internship Completion Certificate



Internship Completion Certificate

Date : 19 May 2026

Sr. No. : IN-BG-VCIP-0526-A792

To Whom It May Concern

This is to certify that **V Sudeep (1CE22CS170)**, a student of **City Engineering College, Bengaluru**, has successfully completed an internship for a period of **02 February 2026 to 16 May 2026** at **Dyashin Technosoft Pvt Ltd., Bengaluru**.

During the internship, he worked extensively on **Python and Data Analytics** and contributed to various projects and tasks.

V Sudeep, demonstrated exceptional skills in Python and Data Analytics and showed a keen interest in learning and applying new technologies. His performance was commendable, and successfully met all the internship requirements and objectives. He also participated in a practical application case study to implement the learned concepts.

We wish him all the best in future endeavors.

Sincerely,

The block contains a handwritten signature in blue ink that reads 'Jasmine'. To the right of the signature is a circular blue stamp. The text around the perimeter of the stamp reads 'DYASHIN TECHNOOSOFT PRIVATE LIMITED' and the word 'Bangalore' is in the center.

Authorised Signatory

ABSTRACT

This report presents a summary of my **4-month internship** at **Dyashin TechnoSoft Pvt Ltd**, where I worked on **Python and Data Analytics**. During the internship, I was involved in tasks such as **data cleaning, preprocessing, exploratory data analysis, and visualization of ecommerce sales data**. Some of the major work carried out includes **ecommerce sales performance analysis, customer behavior analysis, and generating business insights using interactive dashboards**. I applied techniques such as **data analysis, visualization, and statistical analysis** using Python libraries like **Pandas, NumPy, Matplotlib, and Seaborn** to solve problems related to **ecommerce business analysis**. In addition, I gained hands-on experience in **data preprocessing, coding, data interpretation, and report generation**, and used tools like **Python, Microsoft Power BI, Microsoft Excel, Power Query Editor and DAX** to enhance my understanding of practical workflows. This internship helped me improve my **technical skills in data analytics and visualization**, and **soft skills like communication, teamwork, Analytical thinking, problem-solving**, providing valuable exposure to real-world industry practices.

ACKNOWLEDGEMENT

I would like to express our sincere gratitude to everyone who supported and guided us during our **4-month internship** at **Dyashin TechnoSoft Pvt Ltd, J. P. Nagar, Bengaluru, Karnataka**

Firstly, we thank God for His blessings, which have guided us throughout this journey. We also extend our thanks to our institution, **City Engineering College**, for providing the necessary resources and a conducive environment to gain practical knowledge and experience.

Our deepest gratitude goes to **Dr. Thippeswamy H N**, Principal of CEC, for his continuous support and encouragement.

We are also thankful to **Dr. Sowmya**, Head of the Department of Computer Science and Engineering, for her guidance and support during this internship period.

We are grateful to our internship guide, **Mahadeva Prasad H M**, Asst. Prof, Department of Computer Science and Engineering, for his valuable guidance, continuous support, and encouragement throughout the internship.

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A special thank you to our mentor at **Dyashin TechnoSoft Pvt Ltd, Manvith C S**, for providing valuable insights, guidance, and hands-on learning that contributed significantly to our professional development.

We would also like to express our gratitude to our parents and friends for their unwavering moral and emotional support.

Finally, we extend our thanks to all those who contributed their time, knowledge, and resources, making this internship a valuable learning experience.

V SUDEEP(1CE22CS170)

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Chapter 1

INTRODUCTION

Internships play an important role in bridging the gap between academic learning and real-world industry experience. As part of my engineering curriculum, I had the opportunity to undertake a four-month internship at Dyashin TechnoSoft Pvt Ltd, Bengaluru, where I gained practical exposure to Python and Data Analytics. This internship provided me with valuable industry experience and helped me understand how data analysis techniques are applied in real-time business environments.

During the internship, I worked on an Ecommerce Sales Analysis project that focused on analyzing ecommerce sales data to identify customer purchasing behavior, sales trends, order status distribution, revenue generation, payment methods, and product category performance. The project involved handling large datasets and generating meaningful insights to support business decision-making processes.

Throughout the internship, I worked with various technologies and tools such as Python, MySQL, Power BI, Excel, and Jupyter Notebook. I used Python libraries including Pandas, NumPy, Matplotlib, and Seaborn for data cleaning, preprocessing, analysis, and visualization. MySQL was used for managing and querying the database, performing data extraction, and analyzing sales-related information using SQL queries. Power BI was used to create interactive dashboards and visual reports for better understanding of business performance.

The Ecommerce Sales Analysis dashboard included analysis of total revenue, total orders, product category-wise revenue, customer location-based sales, monthly sales trends, payment method distribution, delivery performance, and order status analysis. Through Exploratory Data Analysis (EDA) and visualization techniques, valuable business insights were generated from the dataset.

The internship also enhanced my technical knowledge in data analytics, SQL, and dashboard development while improving my problem-solving, analytical thinking, communication, and teamwork skills. Working in a professional environment helped me gain confidence and exposure to industry-level workflows and practical implementation of analytics concepts.

Internship on Python and Data Analytics

Additionally, the internship helped me understand the importance of data-driven decision making in the ecommerce industry. By analyzing sales patterns and customer behavior, I learned how businesses use analytics to improve product performance, customer satisfaction, marketing strategies, and overall operational efficiency. The project provided practical exposure to interpreting business data and converting raw information into actionable insights through reports and dashboards.

During the development of the project, I also learned the importance of data accuracy, consistency, and proper visualization techniques in analytics. Handling real-world datasets allowed me to understand challenges such as missing values, duplicate records, inconsistent formatting, and data filtering. I gained experience in selecting appropriate charts and visual representations to present complex information in a simple and understandable manner. This practical exposure strengthened my interest in the field of data analytics and enhanced my confidence in working on real-time analytical projects.

This report presents the overview of the organization, details of the training and project work carried out during the internship, the skills and knowledge gained, challenges faced, and the overall learning outcomes achieved during the internship period.

Chapter 2

OBJECTIVE OF INTERNSHIP

The main objectives of the internship are as follows:

- To gain practical knowledge in the field of Data Analytics with Python.
- To understand the real-world applications of data analysis in business environments.
- To develop skills in data cleaning, preprocessing, analysis, and visualization techniques.
- To gain hands-on experience with Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn.
- To understand the process of Exploratory Data Analysis (EDA) using ecommerce sales datasets.
- To learn database management and data querying using MySQL and SQL commands.
- To create interactive dashboards and reports using Power BI and Excel.
- To analyze ecommerce sales data for identifying revenue trends, customer behavior, order status, and product performance.
- To improve problem-solving, analytical thinking, and decision-making skills through practical implementation.
- To understand professional workflows and industry practices followed in data analytics projects.
- To enhance communication, teamwork, presentation, and time management skills in a professional environment.
- To bridge the gap between theoretical knowledge and practical industry experience through real-time project work.
- To gain confidence in handling real-world datasets and generating meaningful business insights.
- To prepare for future career opportunities in the fields of Data Analytics, Business Intelligence, and Data Science.

Chapter 3

COMPANY PROFILE

3.1 Overview of the Organization

Dyashin TechnoSoft Pvt Ltd is an Information Technology and Engineering Services company headquartered in Bengaluru, Karnataka, India. The company focuses on delivering innovative IT solutions, engineering services, and next-generation technology products for various industries. The organization operates in domains such as Information Technology, Semiconductor & Embedded Systems, Mechanical & Automotive Engineering, Artificial Intelligence, Data Analytics, and Digital Transformation solutions.



Fig 3.1.1 Logo of the Company

Dyashin TechnoSoft Pvt Ltd was incorporated in September 2023 and functions as a private limited company registered under the Registrar of Companies, Bangalore. The organization has rapidly expanded its presence and currently serves clients across multiple countries including India, the United States, Australia, and the United Arab Emirates.

The company provides value-driven IT services and focuses on innovation-oriented solutions that help businesses improve efficiency, scalability, and digital growth. Dyashin emphasizes practical implementation, advanced technologies, and industry-oriented workflows to provide reliable and future-ready solutions for modern business challenges.

The organization also supports skill development and technical education through internship programs, industrial training, workshops, and live projects. Through its training initiatives and

practical learning environment, the company helps students and young professionals gain industry exposure and hands-on technical experience.

During the internship period, the organization provided a professional learning environment where practical knowledge and real-time project implementation were given importance. Guidance from mentors and exposure to industry-standard tools helped in understanding the workflow involved in Data Analytics and business intelligence projects.

3.2 Vision and Mission of the Organization

3.2.1 Vision Of The Organization

The vision of Dyashin TechnoSoft Pvt Ltd is to deliver innovative, reliable, and future-ready technology solutions that support digital transformation across industries. The organization aims to become a globally recognized technology partner by combining innovation, quality, and customer-focused services.

3.2.2 Mission Of The Organization

The mission of the organization is:

- To provide value-driven IT and engineering solutions.
- To promote innovation using modern technologies and digital transformation strategies.
- To bridge the gap between academic learning and industrial requirements through practical training and internships.
- To build scalable and reliable solutions for businesses across multiple sectors.
- To encourage continuous learning, teamwork, creativity, and professional development among employees and interns.

3.3 Objective of the Organization

The main objectives of Dyashin TechnoSoft Pvt Ltd are:

- To provide reliable and innovative IT and engineering solutions for various industries.

- To support digital transformation through advanced technologies and analytical approaches.
- To deliver quality software, analytics, and engineering services that satisfy client requirements.
- To provide practical training and internship opportunities for students and professionals.
- To bridge the gap between academic knowledge and industry expectations through real-time project exposure.
- To encourage innovation, research, teamwork, and continuous skill development among employees and interns.

3.4 Roles And Responsibility Of The Organization

The organization follows a structured workflow with different departments and technical teams responsible for various activities.

Technical Team

The technical team handles software development, analytics implementation, testing, and project execution activities. They guide interns and employees in developing practical solutions using industry-standard tools and technologies.

Training and Development Team

This team is responsible for conducting technical training sessions, assigning projects, mentoring interns, and helping students understand practical implementation of concepts.

Management and Administration Team

The administration team manages organizational coordination, project scheduling, communication, documentation, and operational activities within the company.

Quality and Support Team

The quality assurance and support team focuses on maintaining service quality, testing solutions, handling technical issues, and ensuring proper project delivery workflows.

3.5 Organization Structure

Dyashin TechnoSoft Pvt Ltd follows a structured organizational hierarchy to ensure smooth workflow, project execution, training, and management activities. Different departments work together to achieve organizational goals and maintain efficient communication between teams.

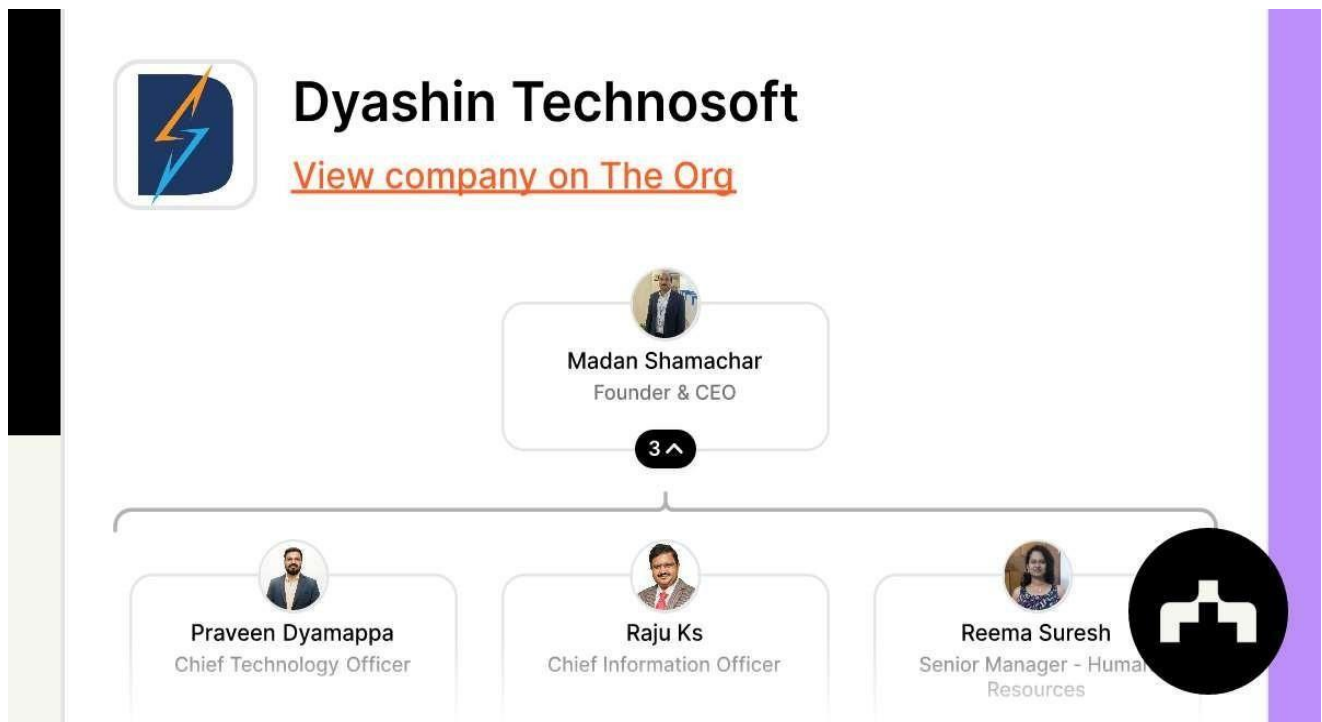


Fig 3.5.1: Organization Structure of the Company



3.5.1 Company Products

Dyashin Technosoft offers a suite of proprietary IT products focused on digital transformation, including MediTrack, DSPristine, Sigo, and Shine Chat. These products are designed for efficiency and innovation across sectors like IT, semiconductor, and automotive, alongside services like DSEdify and DSJobby.

Key Product Portfolio

MediTrack: Likely a tracking or management solution, often tailored for healthcare or asset tracking.

DSPristine: A software solution focusing on process integrity or digital transformation.

Sigo: A product aimed at operational efficiency.

Shine Chat: An enterprise communication or chat application.

Additional Business Verticals

DSEdify: Specialized technology training and educational services.

DSJobby: A recruitment or talent acquisition platform.

Dyashin operates as a software development and engineering services company founded in 2023, with locations in India, the USA, and Australia.

Chapter 4

TRAINING CONTENT

The four-month internship at Dyashin TechnoSoft Pvt Ltd provided me with valuable training in Data Analytics with Python, database management, data visualization, and their real-world business applications. The training content was designed to equip me with the essential technical and analytical skills required to work on real-time data analytics projects and understand practical industry workflows. During the internship, I gained hands-on experience in data preprocessing, exploratory data analysis, SQL querying, dashboard creation, and business insight generation using ecommerce sales datasets. The training was divided into several key areas as outlined below:

4.1 Understanding Company Profile and Ongoing Projects

During the initial phase of the internship at Dyashin TechnoSoft Pvt Ltd, I was introduced to the company profile, organizational workflow, and various technical domains handled by the organization. The company mainly focuses on Information Technology services, Data Analytics, Software Development, Engineering Solutions, and technical training programs. Through the orientation and training sessions, I gained an understanding of the company's work culture, project execution process, and the importance of technology-driven solutions in modern industries.

I was also introduced to different ongoing projects and business applications related to Data Analytics and Business Intelligence. As part of the internship, I worked on an Ecommerce Sales Analysis project, which involved analyzing ecommerce sales datasets to generate meaningful business insights. The project included sales trend analysis, customer behavior analysis, payment method analysis, order status tracking, and dashboard creation using Python, MySQL, and Power BI. This helped me understand the real-world implementation of data analytics concepts and industry-level project workflows.

4.2 Introduction to Programming Languages

During the internship, I was introduced to Python programming, which is one of the most widely used programming languages in the field of Data Analytics and Data Science. The training covered the basic concepts of Python such as variables, data types, operators, conditional statements, loops,

Internship on Python and Data Analytics

functions, lists, tuples, dictionaries, and file handling. These concepts helped in building a strong foundation for performing data analysis and handling real-world datasets efficiently.

I also learned how Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn are used for data preprocessing, manipulation, analysis, and visualization. Along with Python, I gained practical exposure to MySQL and SQL queries for database management and data extraction. SQL commands such as SELECT, WHERE, GROUP BY, ORDER BY, JOIN, and aggregate functions were used to analyze ecommerce sales data and generate meaningful insights. The training helped me understand how programming languages and database technologies work together in real-time analytics projects.

4.3 Mini Project

Title of the Project : E-COMMERCE SALES PERFORMANCE ANALYSIS

The mini project carried out during the internship focused on analyzing ecommerce sales data to understand business performance, customer purchasing behavior, revenue trends, and operational efficiency. The project involved data collection, cleaning, preprocessing, analysis, and visualization using Python, Excel, and, Power BI tools.

The project aimed to generate meaningful business insights from the ecommerce dataset and support data-driven decision making through interactive dashboards and analytical reports.

Objectives of the Project

- To analyze overall sales performance and revenue trends.
- To identify top-performing categories and products.
- To understand customer purchasing behavior.
- To evaluate order status and delivery efficiency.
- To analyze payment method preferences.
- To generate business insights using dashboards and visualizations.

Technologies and Tools Used

- Python
- Microsoft Power BI

- Microsoft Excel
- Power Query Editor
- DAX

Key Analysis Performed:

4.3.1 Data Collection and Understanding

The e-commerce sales dataset was collected and studied to understand customer details, product categories, sales information, order status, delivery operations, and business requirements. Dataset columns, relationships, and business metrics were analysed before dashboard development.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	OrderID	CustomerID	OrderDate	ProductName	Category	SubCategory	Price	Quantity	DiscountPercent	TotalAmount	PaymentMethod	OrderStatus	DeliveryTimeDays	CustomerLocation	Rating
2	ORD00000001	CUS000001	3/9/2024	Cricket Bat	Sports	Equipment	1202.64	3	0	3607.92	Debit Card	Returned		4 Chicago	2
3	ORD00000002	CUS000002	6/26/2025	Bluetooth Headphor	Electronics	Audio	909.62	8	10	6549.26	UPI	Cancelled		1 Melbourne	2
4	ORD00000003	CUS000003	12/21/2025	Wireless Mouse	Electronics	Accessories	1104.83	8	20	7070.91	Net Banking	Returned		1 Denver	3
5	ORD00000004	CUS000004	11/14/2025	Sunglasses	Fashion	Accessories	188.34	2	15	320.18	Debit Card	Cancelled		1	3
6	ORD00000005	CUS000005	5/22/2024	Denim Jeans	Fashion	Apparel	1069.15	3	30	2245.22	UPI	Returned		2 San Diego	2
7	ORD00000006	CUS000006	5/5/2025	Book: Python Basics	Books	Education	1206.47	9	5	10315.32	Wallet	Delivered		2 Miami	4
8	ORD00000007	CUS000007	10/17/2025	Handbag Classic	Fashion	Bags	965.3	8	15	6564.04	Cash on Delivery	Returned		3 Dallas	5
9	ORD00000008	CUS000008	12/20/2024	Smartphone X12	Electronics	Phones	1247.93	9	5	10669.8	UPI	Shipped		2 London	2
10	ORD00000009	CUS000009	5/29/2025	Sunglasses	Fashion	Accessories	954.41	9	10	7730.72	Debit Card	Processing		5 Surat	1
11	ORD00000010	CUS000010	2/16/2024	Stapler	Office	Stationery	1082.35	10	10	9741.15	Wallet	Delivered		14 Los Angeles	1
12	ORD00000011	CUS000011	5/9/2025	Painting Set	Art	Supplies	627.8	1	15	533.63	Cash on Delivery	Returned		3 Vancouver	3
13	ORD00000012	CUS000012	5/27/2024	Science Kit	Education	Toys	94.57	8	15	643.08	Net Banking	Cancelled		2 Manchester	5
14	ORD00000013	CUS000013	5/5/2024	Plant Pot Set	Home	Decor	200.5	6	20	962.4	Wallet	Delivered		6 Seattle	
15	ORD00000014	CUS000014	7/13/2024	LED Desk Lamp	Home	Lighting	1296.92	7	5	8624.52	Debit Card	Processing		5 Abu Dhabi	3
16	ORD00000015	CUS000015	12/27/2024	Shampoo	Beauty	Haircare	748.57	2	15	1272.57	Wallet	Cancelled		Columbus	5
17	ORD00000016	CUS000016	2/16/2025	Textbook: Math	Education	Books	708.53	9	0	6376.77	Credit Card	Cancelled		2 Denver	3
18	ORD00000017	CUS000017	5/1/2024	Guitar Acoustic	Music	Instruments	436.46	7	20	2444.18	Net Banking	Returned		3	5
19	ORD00000018	CUS000018	2/9/2024	Air Purifier	Home	Appliances	1133.67	4	20	3627.74	Wallet	Cancelled		5 Chicago	1
20	ORD00000019	CUS000019	11/2/2024	Desk Organizer	Office	Furniture	1340.23	3	10	3618.62	Wallet	Processing		2 Mumbai	5
21	ORD00000020	CUS000020	12/5/2025	Painting Set	Art	Supplies	453.19	1	10	407.87	UPI	Delivered		7 Brisbane	3
22	ORD00000021	CUS000021	1/24/2025	Notebook Pack	Office	Stationery	868.56	1	20	694.85	Net Banking	Shipped		5 Toronto	
23	ORD00000022	CUS000022	2/24/2025	Smart Watch	Wearables	Fitness	290.22	8	10	2089.58	Net Banking	Shipped		4 Ottawa	2
24	ORD00000023	CUS000023	6/22/2025	Coffee Beans	Food	Beverages	98.11	3	20	235.46	Debit Card	Returned		3 Montreal	4
25	ORD00000024	CUS000024	11/1/2024	Bicycle	Sports	Cycling	534.66	5	20	2138.64	Net Banking	Shipped		2 Boston	5
26	ORD00000025	CUS000025	10/21/2024	Winter Jacket	Fashion	Outerwear	1036.91	8	10	7465.75	Net Banking	Processing		5 Abu Dhabi	3
27	ORD00000026	CUS000026	11/2/2025	Coffee Maker	Home	Appliances	1432.14	3	10	3866.78	Credit Card	Processing		1 Birmingham	4

Fig 4.3.1: Data Collection and Understanding

4.3.2 Data Cleaning and Preprocessing

Data preprocessing was performed using Power Query Editor to improve data quality and consistency. The following operations were carried out:

- Removed duplicate records
- Handled null and missing values
- Renamed columns for better readability
- Corrected inconsistent data formats
- Converted data types
- Removed unnecessary columns
- Replaced missing rating values using median values

This step improved data accuracy and reporting efficiency.

Table.TransformColumnTypes(#"Promoted Headers",{{"order_id", type text}, {"order_date", type date}, {"customer_id", type text},

	order_id	order_date	customer_id	customer_city	customer_state	product_id	product_cat
1	SO-001000	17-08-2022	CUST-0002	Surat	Telangana	PRD-069	Consumable
2	SO-001001	21-02-2022	CUST-0090	Bangalore	Telangana	PRD-096	Raw Material
3	SO-001002	18-07-2023	CUST-UNKNOWN	Pune	Delhi	PRD-017	Raw Material
4	SO-001003	17-05-2023	CUST-0021	Ahmedabad	West Bengal	PRD-098	Consumable
5	SO-001004	15-06-2023	CUST-0005	Kolkata	Maharashtra	PRD-037	Consumable
6	SO-001005	13-10-2022	CUST-0284	Pune	Gujarat	PRD-062	Finished Goods
7	SO-001006	29-07-2022	CUST-0197	Surat	Maharashtra	PRD-090	Packaging
8	SO-001007	28-06-2022	CUST-UNKNOWN	Bangalore	Maharashtra	PRD-031	Raw Material
9	SO-001008	14-05-2024	CUST-0250	Bangalore	West Bengal	PRD-092	Packaging
10	SO-001009	07-03-2022	CUST-0083	Chennai	Maharashtra	PRD-UNKNOWN	Finished Goods
11	SO-001010	03-03-2022	CUST-0236	Mumbai	Delhi	PRD-093	Packaging
12	SO-001011	11-07-2022	CUST-0168	Surat	Telangana	PRD-UNKNOWN	Spare Parts
13	SO-001012	24-03-2023	CUST-0006	Bangalore	Karnataka	PRD-036	Finished Goods
14	SO-001013	15-06-2023	CUST-0142	Mumbai	Karnataka	PRD-026	Spare Parts
15	SO-001014	31-10-2024	CUST-0125	Delhi	Maharashtra	PRD-086	Spare Parts
16	SO-001015	24-02-2022	CUST-UNKNOWN	Delhi	Gujarat	PRD-099	Spare Parts
17	SO-001016	12-02-2023	CUST-0169	Chennai	Tamil Nadu	PRD-028	Consumable
18	SO-001017	09-05-2024	CUST-0024	Bangalore	Maharashtra	PRD-061	Semi-Finished
19	SO-001018	28-03-2023	CUST-0171	Kolkata	Maharashtra	PRD-UNKNOWN	Consumable
20	SO-001019	08-07-2024	CUST-0109	Coimbatore	Tamil Nadu	PRD-050	Finished Goods
21	SO-001020	24-07-2023	CUST-0210	Coimbatore	Delhi	PRD-UNKNOWN	Semi-Finished
22	SO-001021	14-01-2022	CUST-0095	Ahmedabad	Gujarat	PRD-082	Raw Material
23	SO-001022	23-11-2022	CUST-0202	Mumbai	Gujarat	PRD-069	Finished Goods
24	SO-001023	15-05-2024	CUST-0285	Chennai	West Bengal	PRD-093	Semi-Finished
25	SO-001024	15-06-2023	CUST-0002	Ahmedabad	Karnataka	PRD-011	Packaging
26	SO-001025	24-07-2023	CUST-UNKNOWN	Delhi	West Bengal	PRD-088	Consumable
27	SO-001026	15-11-2022	CUST-0231	Ahmedabad	Telangana	PRD-038	Spare Parts

Fig 4.3.2: Data cleaning and Preprocessing

4.3.3 Dashboard Pages and Analysis

The project consists of five interactive dashboards developed in Power BI for analyzing different aspects of e-commerce sales performance.

The dashboards developed in this project are:

- Home Dashboard
- Customer Analysis Dashboard
- Product Performance Dashboard
- Delivery & Operations Dashboard
- Executive Summary & Recommendations Dashboard

4.3.3.1 Home Dashboard

The Home Dashboard acts as the main navigation page of the project. It provides access to all dashboard pages and gives an overview of the complete E-Commerce Sales Analysis project.



Fig 4.3.3.1: Home Dashboard

4.3.3.2 Customer Analysis Dashboard

The Customer Analysis Dashboard focuses on analyzing customer purchasing behavior, sales trends, customer ratings, and order activities. KPI cards such as Total Customers, Total Orders, Average Rating, and Average Order Value were created to provide quick business insights. Various visualizations including Monthly Sales Trend, Sales Distribution by Category, Top Customers by Sales, and Order Status Overview were developed to understand customer engagement and purchasing patterns.

Insights:

- The business maintained a strong customer base with 8K customers generating 20K total orders.
- Monthly sales trends remained relatively stable throughout the analysis period.
- Top customers contributed significantly to overall sales revenue.
- Order statuses remained balanced across different categories.

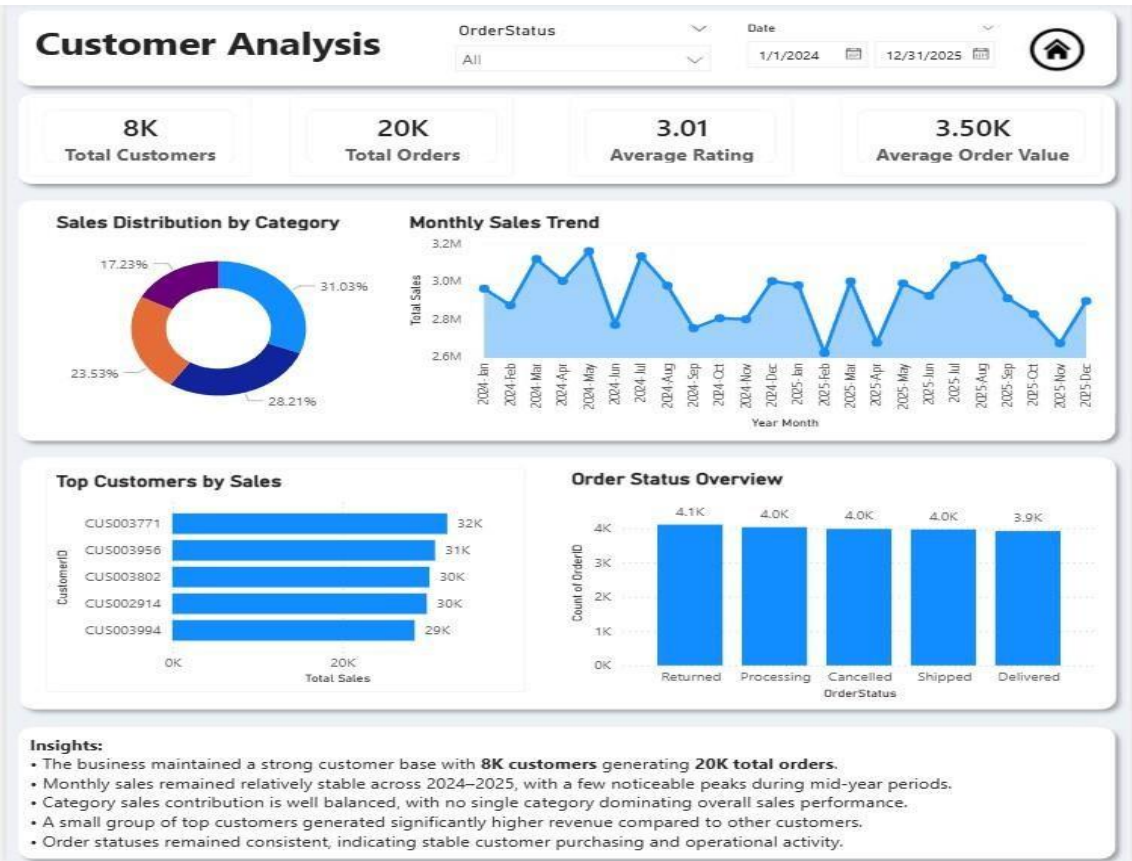


Fig 4.3.3.2: Customer Analysis Dashboard

4.3.3.3 Product Performance Dashboard

The Product Performance Dashboard analyzes sales performance across different product categories. KPI cards including Total Quantity Sold, Total Products, Total Sales, and Top Product Quantity were used to evaluate overall product performance. Charts such as Top Products by Revenue, Category-wise Sales Distribution, Category Performance Summary, and Tree map Analysis were created to identify high-performing products and category-wise sales contribution.

Insights:

- The business generated 70.07M in total sales with over 109K units sold.
- Office category contributed the highest share of sales revenue.
- Products such as Vitamin D3, Winter Jacket, and Yoga Mat emerged as top-performing products.
- Product ratings remained stable across multiple categories.

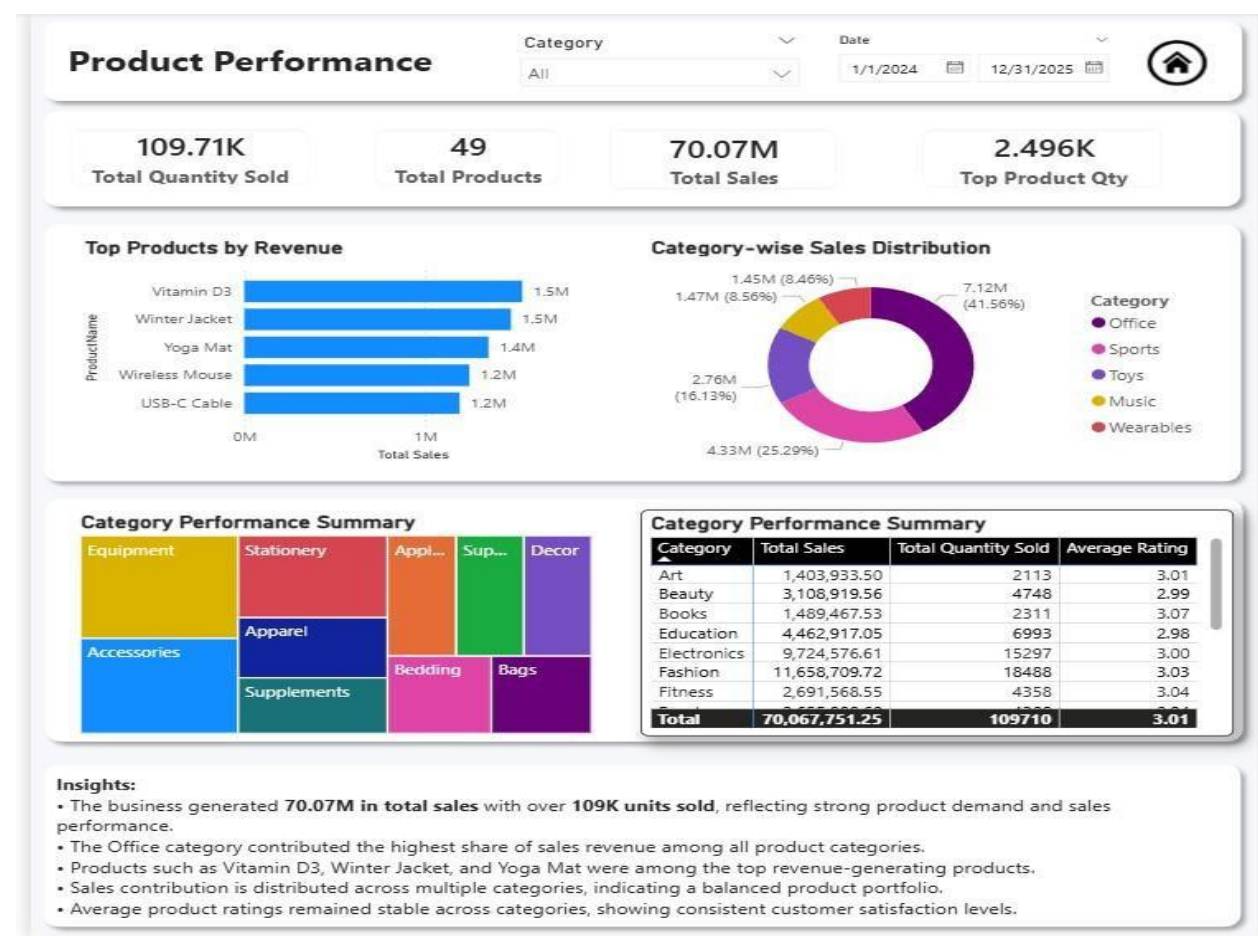


Fig 4.3.3.3: Product Performance Dashboard

4.3.3.4 Delivery & Operations Dashboard

The Delivery & Operations Dashboard focuses on operational performance and order delivery analysis. KPI cards such as Total Orders, Delivered Orders, Returned Orders, and Cancelled Orders were used to monitor delivery activities and operational efficiency. Visualizations including Monthly Order Trend, Category-wise Returned Orders, Order Status by Category, and Location-wise Operational Summary were created to analyze delivery performance and operational activities across multiple customer locations.

Insights:

- Approximately 4K orders were successfully delivered during the analysis period.
- Returned and cancelled orders highlighted operational improvement opportunities.
- Home and Fashion categories recorded higher returned orders compared to other categories.
- Operational performance remained relatively stable across customer locations.



Fig 4.3.3.4: Delivery & Operations Dashboard

4.3.3.5 Executive Summary & Recommendations Dashboard

The Executive Summary & Recommendations Dashboard summarizes the major business insights, sales trends, operational observations, and recommendations identified from the dashboard analysis. This dashboard provides a consolidated overview of business performance and improvement strategies.

The dashboard includes key business insights, operational recommendations, customer retention strategies, sales improvement strategies, major trends, and assumptions considered during analysis.

Key Insights:

- The business generated over 70M in total sales with strong customer engagement.
- Monthly sales trends remained stable throughout 2024–2025.
- Office category emerged as the highest revenue contributor.
- Customer retention and operational efficiency require continuous monitoring.

Recommendations:

- Improve delivery efficiency to reduce returned and cancelled orders.
- Strengthen product quality checks in high-return categories.
- Focus promotional campaigns on top-performing products and categories.
- Improve customer support and customer retention strategies.

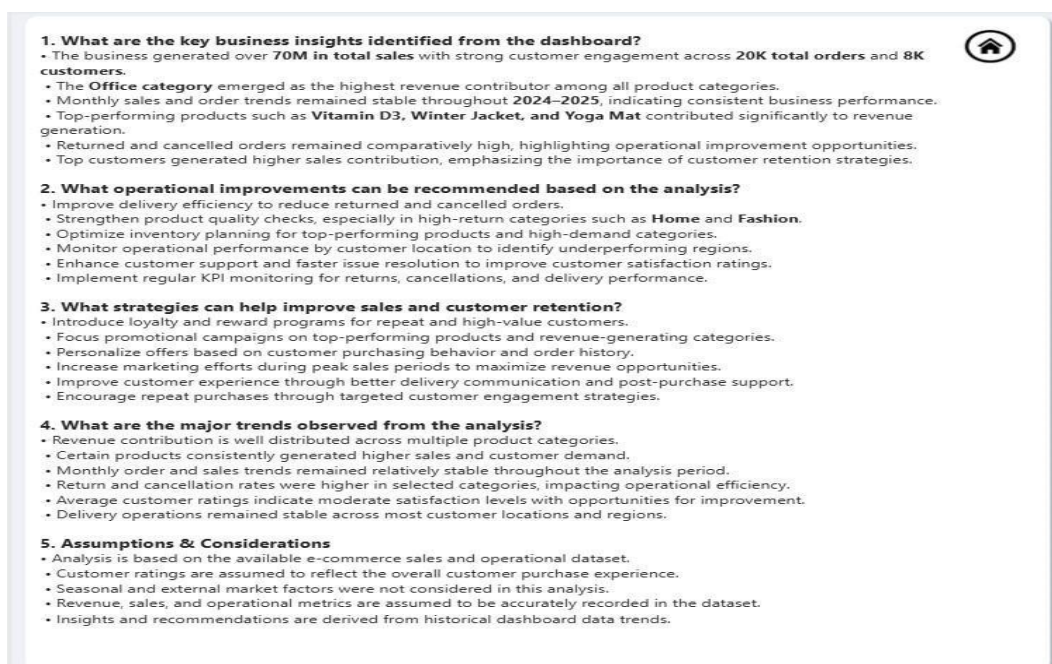


Fig 4.3.3.5: Executive Summary & Recommendations Dashboard

4.4 Tools/Software Used

Various tools and software were used during the internship for data analysis, database management, visualization, and report generation. These tools helped in performing efficient data processing, analysis, and dashboard creation for the Ecommerce Sales Analysis project.

Python

Python was used as the primary programming language for data analysis and preprocessing tasks. It was used for handling datasets, performing calculations, and generating analytical insights.

Microsoft Power BI

Used to create interactive dashboards, visualizations, KPI cards, and reports for analyzing e-commerce sales performance and business insights.

Microsoft Excel

Used for storing the dataset, organizing sales records, and performing basic data handling before importing the data into Power BI.

Power Query Editor

Used for data cleaning, removing duplicates, handling missing values, transforming data formats, and preparing the dataset for analysis.

DAX

Used for creating calculated measures, KPIs, custom columns, and performing analytical calculations within Power BI dashboards.

4.5 Problem Statements

In the rapidly growing ecommerce industry, businesses generate large volumes of sales data every day. However, without proper analysis, it becomes difficult for organizations to understand customer behavior, sales performance, revenue trends, and operational efficiency. The lack of meaningful insights from raw data can affect business decision-making, customer satisfaction, and overall growth.

The main problem addressed in this project was to analyze ecommerce sales data and generate useful business insights that could help improve decision-making processes. The project focused on identifying sales trends, top-performing categories, customer purchasing patterns, payment preferences, delivery performance, and order status distribution using data analytics techniques.

Chapter 5

LEARNING OUTCOMES

The four-month internship at Dyashin TechnoSoft Pvt Ltd has been an enriching and valuable learning experience, providing me with practical knowledge in both technical and professional domains. This chapter highlights the major learning outcomes gained during the internship, focusing on the technical skills acquired in Data Analytics with Python, MySQL, Power BI, and data visualization, as well as the professional and soft skills developed throughout the internship period. The internship also provided exposure to real-world business analytics workflows and enhanced my understanding of how data-driven decision-making is implemented in the ecommerce industry.

5.1 Technical Learning Outcomes

5.1.1 Technical Skills Acquired

During the internship, I developed various technical skills related to programming, database management, data analysis, and visualization.

Python Programming

- Improved understanding of Python programming concepts.
- Learned practical implementation of Python for data analytics tasks.
- Worked with Python libraries for preprocessing and visualization.

Data Analysis and Visualization

- Learned Exploratory Data Analysis (EDA) techniques.
- Performed data cleaning and preprocessing.
- Created charts, graphs, and dashboards for business insights.
- Understood trend analysis and data interpretation.

Database Management using MySQL

- Gained practical exposure to MySQL database management.
- Learned SQL queries for data extraction and analysis.

Internship on Python and Data Analytics

- Used filtering, grouping, sorting, and aggregation operations.

Dashboard Development

- Developed interactive dashboards using Power BI.
- Represented sales performance and business insights visually.
- Improved understanding of Business Intelligence reporting.

5.1.2 Analytics and Problem-Solving Skills

The internship improved my ability to analyze real-world datasets and generate meaningful insights.

Business Insight Generation

- Learned how to analyze ecommerce sales data effectively.
- Identified sales trends and customer purchasing behavior.
- Understood revenue analysis and performance tracking.

Problem-Solving Ability

- Improved analytical thinking and logical reasoning.
- Learned how to handle missing values and inconsistent data.
- Developed the ability to solve data-related problems efficiently

5.2 Soft Skills and Professional Development

The internship also helped in enhancing communication and professional skills required in industry environments.

5.2.1 Teamwork and Collaboration

- Worked collaboratively during project activities and discussions.
- Learned the importance of coordination and teamwork in project execution.

5.2.2 Problem-Solving and Critical Thinking

- Improved analytical thinking and logical reasoning.
- Learned how to handle missing values and inconsistent data.
- Developed the ability to solve data-related problems efficiently.

5.2.3 Time Management

- Learned how to manage project tasks and complete work within deadlines.
- Improved the ability to balance learning and practical implementation.

5.2.4 Communication and Presentation Skills

- Improved technical communication and presentation skills.
- Learned to explain analytical results clearly using visual reports and dashboards.
- Gained exposure to real-world industry workflows and project environments.
- Understood professional ethics and organizational work culture.

Chapter 6

CONCLUSION

The four-month internship at Dyashin Technosoft Pvt Ltd was a valuable and enriching learning experience that provided practical exposure to the field of Data Analytics and Business Intelligence. The internship helped bridge the gap between theoretical knowledge and real-world business applications, improving both technical and professional skills.

During the internship, I worked on the **E-Commerce Sales Performance Analysis** project, which focused on analyzing e-commerce sales data to understand customer purchasing behavior, sales trends, revenue generation, delivery performance, and payment method preferences. The project involved data collection, preprocessing, cleaning, analysis, and dashboard development using Microsoft Power BI and Microsoft Excel. This practical implementation helped me understand the importance of data-driven decision-making in modern business environments.

Through this project, I gained hands-on experience in data cleaning, transformation, visualization, KPI analysis, and dashboard development. I also improved my understanding of business analytics workflows, including data preprocessing, exploratory data analysis, visualization techniques, and report generation.

The interactive dashboards provided valuable insights into product performance, customer behavior, sales trends, and operational efficiency. The internship improved my analytical thinking, problem-solving, communication, and teamwork skills. Working with real-time datasets enhanced my confidence in handling data analytics tasks. It also helped me understand how business insights support decision-making.

Overall, the internship experience significantly contributed to my technical growth and professional development. The knowledge and practical exposure gained during this internship will be highly beneficial for my future academic projects and career opportunities in the fields of Data Analytics, Business Intelligence, and Data Science.

BIBLIOGRAPHY

1. Chandan Gowda, Python for Data Analysis, https://youtube.com/playlist?list=PLlGueSbLhZoBRnTsGiDJeTXuQCALOTN07&si=_oagafq3SiS_1j8l0
2. W3School Python Tutorial, <https://www.w3schools.com/python/>
3. Pavan Lalwani, Power BI tool, <https://youtu.be/2FIVADp6PCw?si=AjWiw8vv5PZUMI3c>
4. Apna College, MySQL, https://youtu.be/hlGoQC332VM?si=cOaC_1Yq0-FcoZMO
5. Abhijeet Pathak, Dashboard reference <https://www.linkedin.com/in/abhijeetpathak23/>

DECLARATION

I hereby declare that the Internship titled "**Internship on Python and Data Analytics**" is an original work carried out by me under the supervision of Mr. **Mahadeva Prasad H M, Asst-Prof, Dept of CSE**, and it has not been submitted anywhere else for any academic degree or certification.

I also certify that all sources of information and data used in the project have been appropriately acknowledged and referenced. Any work done by others has been properly cited, and all assistance received during the Internship has been clearly stated.

I take full responsibility for the integrity and originality of the content in this Internship report.

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